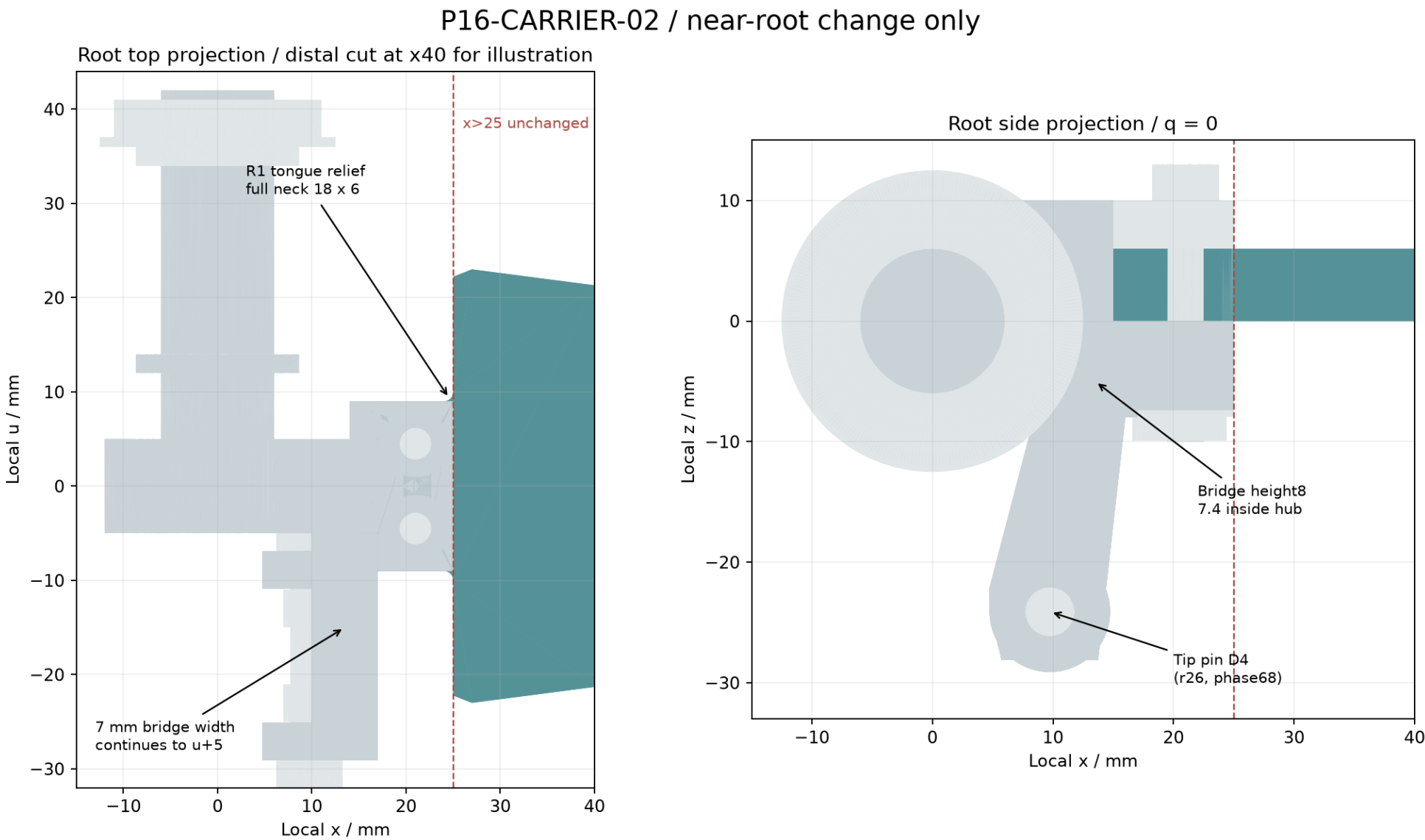


1 mm nominal root clearance without thinning the tongue



Authoritative original STEP / local frame
UR / LL metal_blade_with_narrow_tongue
UR / LL integral_steel_crank_spindle

x: petal length; u: hinge axis; z: face.
Tongue x15..24: u +/-9; thickness6.
R1 arcs from (24,+/-9) to (25,+/-10).
Two D3.4 holes: x21, u +/-4.5.
Distal x>25: exact original geometry.

Steel bridge: x10..17.
u -29.1..-5: z -8..0 (7 x 8).
u -5..+5: z -7.4..0 (7 x 7.4).
Minimum inscribed path: 51.8 mm².
Hub overlap10; avoids M2 keeper heads.
All old shaft/pin datums retained.

Unchanged nominal bearing interface
Solid shaft D12; centers u19 and29.
P16 actuation plane u-18.
Root R70; upper z0 / lower z+50.
Upper q0..109 / lower q0..122 deg.

The OD48 coaxial conservative support shell has exactly 1.000 mm nominal gap for all four roots.

This is also the geometric ceiling at the immutable distal x25 face; tolerances and elastic motion remain unqualified.

Clearance is an analytic rotational-invariance result, not an end-state screenshot result.

Steel material was added; the 18 x 6 tongue was not thinned. Four independent hinge axes and mirrored modules remain.

J7 contact stays head z-140; face z0. Parent 24 mm wrist extension is not integrated. Original wrist +/-90 remains blocked.

Loads use the real offset and the net section after holes

Geometric reaction screen / local u (not pressure centers)



300 N at u-18, bearings at u19 / u29:
|RA| = 4.7 F = 1410 N; |RB| = 3.7 F = 1110 N.
Offset alone: 300 x 18 = 5400 Nmm.
Additional opposed pair due to offset: 540 N.
Actuator-only hinge moment needs a stop or inertia;
ideal radial bearings do not supply hinge torque.

Upper / diameter80, mu0.4, 2kg x factor2, one free-sharing LP witness

eta0.85 + present bare-finger gravity: F 211.2 N; radial reactions 983.3 / 778.4 N.
300 N plus same contacts: 1399.9 / 1106.5 N; unbalanced hinge torque is explicitly retained.

Lower / diameter80, mu0.4, 2kg x factor2, one free-sharing LP witness

eta0.85 + present bare-finger gravity: F 119.3 N; radial reactions 550.1 / 435.9 N.
300 N plus same contacts: 1396.9 / 1103.3 N; unbalanced hinge torque is explicitly retained.

Modeled mass 2.747 kg; full head planning 2.982..3.267 kg, excluding J7 and parent extension.
Bare finger gravity omits later real lamps and pad retainers; LP witness is not a load-sharing hardware result.
M3 preload, clamp separation, hole/fillet fatigue, true bearing pressure centers, P16 duty/holding and cable motion remain open.
Reproduce: p16_carrier02_study.py --motion; p16_carrier02_loads.py; p16_carrier02_export.py; draw_p16_carrier02.py.
Sources and exact arrays: geometry.json, motion.json, loads.json, qa-mass.json; see docs/engineering/p16-carrier02-study.md.

Net-section areas from exact CAD slices
Bridge retained sub-prism: 7 x 7.4 = 51.8 mm2.
Main shaft at u11.9: D12 = 113.097 mm2.
Aluminum tongue at M3 row: 67.200 mm2.
Steel fork at nut pockets: 54.826 mm2.
Upper clamp cap at M3 row: 44.800 mm2.

300 N bridge directional conservative screen:
Bending upper 88.1 MPa; torsion shear 82.5 MPa.
Combined nominal upper 180.9 MPa.
Factor2 requires yield >= 361.9 MPa BEFORE concentrations.
Actual corner/fillet stress and fatigue remain open.
A material name alone does not close this gate.

Steel candidate: original D80 x L80 bar, +QT.
Ovako6082/MoC410M, 40<D<100: Rel >=650 MPa.
Certificate/heat state required; notch/fatigue still open.